LP-Eval: Rubric and Dataset for Measuring the Quality of Legal Proposition Generation

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Abstract

Legal proposition generation is central to legal reasoning and doctrinal scholarship, yet remain under-examined in Legal NLP. This paper investigates the automatic generation and evaluation of legal propositions from decisions of the Court of Justice of the European Union using large language models (LLMs). We introduce LP-Eval, a three-step evaluation rubric co-designed with legal experts that decomposes legal proposition quality into formal validity and substantive dimensions. Using this rubric, we release a dataset of two experts' annotations for 100 LLM-generated legal propositions. Our results show that LLMs can generate predominantly well-formed and high-quality propositions, while expert evaluations reveal higher quality for propositions derived from well established cases than from recent ones. We further examine LLMs as evaluators and find that rubric-guided LLM judgments align more closely with expert assessments than direct overall scoring, but remain insensitive to finer-grained distinctions captured by human experts.

CCS Concepts

• **Applied computing** → **Law**; • **Computing methodologies** → **Natural language generation**; **Language resources**.

Keywords

Natural Legal Language Processing, Large Language Models, Legal Proposition, Evaluation Rubric

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1 Introduction

Legal work, whether in public administration, private corporations, solicitors or barristers offices or courts, centers around determining what the law 'is'. Only if we know what the law is, can we apply it to the case we have before us. Lawyers, then, constantly have to state what the law is. Such statements, often called statements on 'valid law', articulate the content of general legal rules and principles in some jurisdiction. Such claims, which we in this paper refer to as *Legal Propositions* (LPs), are statements on legal norms (rules or principles, i.e. they are normative and not factual statements), related to their scope, conditions, or consequences. LPs are foundational to doctrinal legal scholarship: the central aim of legal textbooks and legal commentary (often published as journal articles) is to state what the law is by formulating and refining such propositions.

Large-scale collections of legal propositions would have substantial practical value. Legal propositions are lawyers' primary unit of legal normativity; without them, legal analysis and knowledge organization become markedly more difficult. Legal propositions are today still exclusively manually constructed, either by legal academics when they write textbooks or legal commentary or by legal practitioners when they write pleas or decisions. For example, in the U.S., leading providers of legal information use human editors to generate legal propositions for court decisions, which they refer to as Headnotes [10].

Automating legal proposition generation, the task of producing legal statements based on case decision paragraphs and citation contexts (see upper panel of Fig 1), has the potential to improve both the coverage and quality of a wide range of LegalTech applications and downstream legal reasoning pipelines. Despite its practical importance, automatic legal proposition generation has received comparatively limited attention in prior Legal NLP and LegalTech research. Much prior work on legal reasoning has focused either on the sources of law (legislation and prior decisions) e.g., precedent cases retrieval [27], based on the narrative of case fact (e.g., judgment outcome and related classification tasks [5]. Legal proposition generation is complementary to these lines of work: it targets the interface between "raw" legal sources and the fact patterns of new disputes by producing normative statements that can be invoked, combined, and applied in subsequent analysis. As such, it sits at the *nexus* between legal authority and factual

Automatic Legal Proposition Generation

Paragraph from CJEU Decision: *That article therefore lays down strict conditions for the adaptation, by the competent authority of the executing State, ... under Article 8(1) of that framework decision, to recognise a judgment which has been forwarded to it and to execute the sentence ...*
Citation Paragraphs: *In particular, Article 8 of Framework Decision 2008/909 ...*

Legal Proposition generated by GPT_{oss}: Article 8(1) of Framework Decision 2008/909 imposes a general duty on the executing State to recognise and execute a forwarded judgment ...

Expert Annotation based on LN-Eval Rubric

Formal Components: Stance ✓ Object ✓ Specification ✓ *(VALID)*

Quality Dimensions:	Source Independence	2/3
	Fact Independence	3/3
	Conciseness	3/3
	Generality	3/3
	Fidelity	3/3

Overall: 3/3 *(Excellent)*

Figure 1: Example of legal proposition generation and expert annotation based on LN-Eval rubric

context, offering a potentially useful intermediate representation for both human and machine legal reasoning.

A key challenge in legal proposition generation is evaluation. As an open-ended task, there is rarely a single "gold" reference: multiple outputs may be valid, while small wording differences can alter legal meaning. Consequently, n-gram–based metrics such as BLEU [22] and ROUGE [19] are ill-suited, as they capture surface overlap rather than legal content. Another challenge is grounding: propositions must be faithful to authoritative sources. Recent work explores *LLM-as-judge* for evaluation [17], but this approach remains vulnerable to hallucinations [18], such as accepting unsupported claims or fabricated citations [8].

These concerns have motivated recent NLP work in rubric-based evaluation protocols that decompose quality into human-designed explicit, structured criteria (e.g., faithfulness and conciseness) [14], rather than a single holistic score. For instance, TN-Eval [24] develops a rubric and associated protocols to assess the quality of behavioral therapy notes and reports improved reliability and interpretability compared to traditional evaluation approaches.

Motivated by these challenges, we investigate the ability of LLMs to generate and evaluate legal propositions from decisions of the Court of Justice of the European Union (CJEU). To our knowledge, this is the first study in legal NLP to systematically examine both the automatic generation and evaluation of legal propositions. Our contributions are:[1] 1) We introduce LP-Eval, a three-step rubric co-designed with legal experts to support structured and consistent evaluation of generated legal propositions (illustrated in Fig 2). 2) We release the LP-Eval dataset, consisting of expert annotations for LLM-generated legal propositions following the LP-Eval guidelines. 3) Through quantitative experiments, we show that LLMs can act as both generators and evaluators of legal propositions, though with notable limitations: experts rate propositions from well established cases higher than those from recent cases, while LLM-based evaluators, despite improved alignment when using LP-Eval, fail to capture such finer-grained distinctions.

2 Task Definition

We examine LLMs' ability to generate and evaluate LPs using decisions of the CJEU. Our experiments comprise two tasks: *LLM as generator* and *LLM as evaluator*.

LLM as generator: We prompt off-the-shelf LLMs to generate LPs using an expert-designed template that provides instructions of the task, definition of LP, focus paragraph, and its cited context. § 4.2 offeres details of the prompt and implementation. Two legal experts evaluate the generated propositions using the LP-Eval rubric (§ 4.3), which we treat as ground truth for the generation quality.

LLM as evaluator: We prompt LLMs to evaluate generated propositions using a template derived from the LP-Eval rubric. Evaluator performance is assessed by measuring agreement between LLM scores and expert annotations. Details of the evaluating prompt and implementation are given in § 4.4.

3 LP-Eval Rubric

This section introduces our rubric for evaluating generated LPs. Due to space constraints, we focus on design decisions; a full description is provided in App C.

Building on prior work in rubric design [9, 12], our rubric comprises two elements: required LP **COMPONENTS** and quality **DIMENSIONS** distinguishing stronger from weaker outputs.

We developed the rubric through a two-step co-design process with legal experts. First, interviews with two European law professors identified key components and dimensions, which were iteratively refined and summarized into an initial draft. Second, experts applied the rubric to a sample of LLM-generated LPs, followed by feedback to assess its discriminative ability and further refine it. Finally, a computational linguist formalized the evaluation protocol, which we used to develop expert annotation guidelines for human evaluation and a prompt template for LLM-as-judge automatic evaluation.

4 LP-Eval Dataset

The dataset consists of 50 paragraphs sampled from 10 CJEU decisions. For each paragraph, we include two LLM-generated LPs, manual annotations on the quality of the generated LPs from two legal professionals following the LP-Eval rubric, and corresponding automatic evaluations produced by GPT-oss[2], an open LLM.

4.1 CJEU Cases Sampling

We sampled the 10 CJEU decisions from a proprietary database compiled and curated by LawLibrary.AI[2] from different official sources. Previous work reveals that LLMs to tend memorize parts of their training data [4]. A possible concern is therefore that LLMs

[1]Our dataset and code will be released publicly upon acceptance.

[2]https://www.lawlibrary.ai/

may generated LPs based on their memorisation of the pretraining data. This is particularly likely when it comes to older and more frequently discussed court decisions. To test for such effects, we selected a combination of *well-established* and *recent* decisions. The *well-established* group consists of five highly-cited and discussed decisions, spread across time that were manually selected by the legal experts (Full list of the selected cases are available in App D). We selected five paragraphs per case, resulting in a dataset of 50 paragraphs (see App D for sampling details). Descriptive statistics are reported in § 4.5 in Appendix.

4.2 LLM Legal Proposition Generation

We employed three models for proposition generation: (1) GPT-OSS: A large-scale open-source model (120 billion parameters) [2]. (2) OLMo-3-7B-Instruct: A 7 billion parameter instruction-tuned model [21]. (3) Saul-7B-Instruct: A 7 billion parameter model specifically trained on legal data [7]. All models were configured with a temperature of 0.3 to balance between determinism and output diversity. See App A for implementation details and expert-crafted LP generation prompts.

4.3 Expert Evaluation Collection

We recruited a final-year law student and a research assistant with a PhD in law to do the annotation. Both were paid current collective bargaining salaries through their respective universities. They conducted annotations following the LP-Eval annotation protocol. Due to resource constraints, human evaluation was limited to the LPs generated by GPT-OSS and OLMo3, which demonstrated the most promising quality in an initial expert assessment. The annotation process was done on the label studio platform [26].

Inter-Annotator Agreement We observe a strong positive skew in the experts' ratings. (Fig 3 presents the distribution of the annotation results). Only 5 out of 200 generated LPs are labeled as **INVALID**, due to missing any one of the required formality components (i.e., specification, object, and stance). Across the three quality dimensions rated on a 1–3 Likert scale, the annotation distributions exhibit substantial class imbalance, with the highest rating ("3") assigned to the majority of items in all dimensions.

Given the pronounced class imbalance, we assess inter-annotator agreement using **Gwet's AC1** [13], consistent with prior work [3]. We prefer AC1 over the more common Cohen's Kappa due to its robustness against the *prevalence paradox* [11], which can artificially deflate reliability estimates in skewed distributions despite high observed agreement. We obtained a high agreement score (AC1 = 0.96), indicating strong consistency among annotators and validating the clarity of the LP-Eval Rubric guidelines.

4.4 Automatic LLM Evaluation

We also assess proposition quality via an LLM-as-a-judge approach, utilizing GPT-OSS, OLMo3, and Saul as evaluators. All evaluator models were configured with a temperature of 0 to ensure deterministic behavior and maximize reproducibility. The evaluation prompts were constructed using templates derived from the LN-Eval Rubric (see App C in Appendix).

4.5 Dataset Statistics

Tab 4 shows that paragraph lengths are comparable across categories, with *recent* cases averaging 174.56 tokens and *well-established* cases 159.24 tokens (std. 50.75 and 68.79). Generated LPs are substantially shorter than their source paragraphs for both models: GPT-OSS produces LPs averaging 48.92 tokens, while OLMo-3 produces slightly longer LPs at 51.98 tokens, corresponding to a compression of approximately 29–31% relative to the average paragraph length (166.90 tokens). The length of generated LPs also varies by model and case category. OLMo-3 generates longer LPs than GPT-OSS for *recent* cases (59.12 vs. 52.36 tokens), while their outputs are nearly identical for *well-established* cases (44.84 vs. 45.48). Saul-LM produces slightly longer LPs (53.30 tokens) than GPT-OSS (48.92) and is close to OLMo-3 (51.98). The category trend is consistent: Saul-LM generates longer LPs for *recent* cases (59.84) than for *well-established* cases (46.76). Notably, Saul-LM shows higher variance in LP length, especially for *recent* cases, which suggests less consistent compression behavior than GPT-OSS or OLMo-3.

5 Experiments

5.1 LLM as LP generator

Expert Evaluation of LLM-Generated Legal Propositions
We evaluate the quality of LLM-generated LPs through expert assessment following our LP-Eval rubric. An LP is considered valid only if it contains all three required components: *Stance*, *Object*, and *Specification*. Across the dataset, experts judged the vast majority of generated LPs to be formally valid. Out of 100 LPs, 95 were rated as VALID. At the formal **COMPONENT** level, experts consistently identified the presence of both *Stance* and *Specification* in all generated LPs. The five invalid cases all failed due to a missing *Object* component, defined as a normative (rather than factual) statement. Furthermore, *Object* component was a significant source of inter-annotator disagreement. According to the qualitative study from legal experts (see App F), legal norms consist of different subtypes, which may have confused the expert annotators.

Tab 2 summarizes expert evaluations at Step 2 (quality **DIMENSIONS**) and Step 3 (**OVERALL** quality scores). Experts assigned consistently high scores for *Fact Independence* (mean = 2.96) and *Fidelity* (mean = 2.95), indicating that the generated propositions are largely abstracted from case-specific facts and are well supported by the focus paragraph and its surrounding context. *Conciseness* and *source independence* received slightly lower but still positive average scores (means = 2.48 and 2.31, respectively), reflecting occasional redundancy or explicit references to legal sources. The **OVERALL** quality score averaged 2.5, rated as high on our 3-point Likert scale. Valid propositions frequently received ratings of 3.0 ("Excellent") or 2.0 ("Satisfactory"), suggesting that while experts prefer more source-independent drafting, they generally regard the generated propositions as competent and legally sound summaries of the underlying norms.

Quality Difference in Well-Established vs. Recent Cases
We compared expert evaluation scores for generated LPs derived from *well-established* cases against those from *recent* cases. For each quality dimension, we computed the mean expert score per case-prominence group ($\text{established}_{\text{meann}}$ vs. $\text{recent}_{\text{mean}}$) and performed paired two-sample t-tests to assess differences between the two

Case Categories	token_length: paragraph text	token_length: GPT generated LP	token_length: OLMO generated LP	token_length: Saul-LM generated LP
recent cases	174.56 ± 50.75	52.36 ± 11.86	59.12 ± 14.60	59.84 ± 25.94
well established cases	159.24 ± 68.79	45.48 ± 13.02	44.84 ± 13.66	46.76 ± 16.36
overall	166.90 ± 60.32	48.92 ± 12.81	51.98 ± 15.74	53.30 ± 22.46

Table 1: Dataset Statistics of the 100 selected CJEU case text paragraphs and generated LPs (mean ± std)

fact_ind	fidelity	concise	src_ind	general	overall
2.96 ± 0.19	2.95 ± 0.23	2.48 ± 0.62	2.31 ± 0.63	2.65 ± 0.64	2.50 ± 0.60

Table 2: mean and avg of experts annotation. Likert score 1-3

	$\text{recent}_{\text{mean}}$	$\text{established}_{\text{mean}}$	t_statistic	p_value
fact_ind	2.95	2.98	-1.13	0.26
fidelity	2.94	2.97	-0.91	0.37
general	2.57	2.73	-1.77	0.08
concise	2.38	2.58	-2.27	2e-02 **
src_ind	2.11	2.52	-4.74	4e-06 **
overall	2.35	2.66	-3.79	2e-04 **

Table 3: T-tests comparing LPs generated from *well established* cases to *recent* cases

groups. As reported in Tab 3, statistically significant differences emerge for *Conciseness*, *Source Independence*, and **OVERALL** Quality. LPs generated from *well-established* cases were rated as more concise ($p = 0.02$) and more source-independent ($p < 10^{-5}$). These advantages are reflected in the **OVERALL** quality scores, which are significantly higher for *well-established* cases (mean = 2.66 vs. 2.35, $p < 0.001$). These results suggest that while core properties, such as factual abstraction and contextual alignment, are stable across case types, higher-level qualities related to abstraction and formulation vary with case prominence. A qualitative analysis by legal experts (App F) suggests these disparities may stem from the variations in inherent structural differences between the case categories.

5.2 LLM as LP Evaluator

Alignment Between LLM and Expert Evaluations

Among the three LLMs tested, OLMo and SaulLM assigned maximum scores to all propositions (zero variance), therefore we focus on the analysis of GPT-OSS's evaluation. GPT-OSS achieves Gwet's AC1 agreement of 0.91 and 0.93 with the two experts. While slightly lower than the inter-expert agreement 0.94, these values indicate comparable agreement when GPT-OSS follows the LP-Eval protocol. To test the LN-Eval rubric's efficacy, we prompted GPT-OSS in a *one-go* setting: evaluate only the **OVERALL** quality score, skipping all intermediate rubric steps for formal COMPONENT and quality Dimensions (see § C.1). Agreement drops to 0.85 (approximately 7% relative decrease), demonstrating that structured rubric guidance substantially improves alignment with expert judgments.

LLM-Judge Sensitivity to Case Prominence We conducted a t-test comparing GPT-OSS evaluations of LPs derived from *well-established* and *recent* cases but found no statistically significant differences between the two groups. In contrast, expert evaluators identified statistically significant distinctions across these case categories. This suggests that while GPT-OSS broadly aligns with experts at an overall level, it is less sensitive to the nuanced qualitative differences captured by human experts.

6 Related Work

Legal NLP Recent work in computational law has explored various Legal NLP tasks such as legal case retrieval [20], judgement prediction [28] , legal case summarization [1], vulnerability detection [31], and drafting support for legislative text [6]. Across these research lines, proposition-level representations remain largely implicit in statute-based formulations [15] and are only partially captured in case-law settings [23]. Our work targets this by focusing on proposition-level representations derived from case law and assessing whether language models can generate and preserve the normative propositional content that supports downstream legal reasoning.

LLM-as-a-Judge is increasingly used as a scalable alternative to expert human assessment for generation tasks, but its reliability depends heavily on the evaluation protocol and prompt design [34]. Recent analyses show that zero-shot LLM evaluators can exhibit systematic biases [25]. To reduce ambiguity and improve reproducibility, recent work emphasizes more explicit rubric construction. CheckEval decomposes subjective criteria into Boolean checklist questions, improving reliability relative to single scalar ratings, improving inter-evaluator agreement [16].

7 Limitations

Due to computational constraints, we do not perform a causal analysis of pre-training data and its influence on model outputs. Prior work shows that LLMs can reflect biases in their training data [30], motivating future work on detecting and mitigating memorized biases. Owing to space limitations, we emphasize quantitative analysis, with detailed qualitative results provided in App F.

Finally, while expert annotations show high overall agreement, disagreements persist on abstract components (e.g., *object*). This is consistent with prior findings on limited inter-expert agreement in legal NLP [33] and imperfect alignment between models and inherently diverse human judgments [32]. Given the high-stakes legal setting, future work should better incorporate annotation variability to preserve pluralistic human values and support human-centered AI systems [29].

8 Conclusion

We investigate the automatic generation and evaluation of legal propositions (LPs), a core yet underexplored unit of legal reasoning. In collaboration with legal experts, we introduce LP-Eval, a rubric-based evaluation framework for assessing both the validity and quality of generated LPs, along with corresponding evaluation protocols for both expert annotation and LLM-as-judge evaluation. We further release the LP-Eval dataset, a curated corpus of CJEU

cases and LLM-generated legal propositions annotated by two legal experts following the LP-Eval rubric. Our experiment results show that modern LLMs can generate generally well-formed and high-quality legal propositions, while exhibiting systematic quality differences across case-prominence types. We also find that rubric-guided LLM evaluation aligns more closely with expert judgments than direct overall scoring, but remains insufficiently sensitive to finer-grained distinctions captured by human evaluators. These findings highlight both the promise and current limitations of LLMs for legal proposition generation and evaluation.

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A Implementation Details

Our proposition generation pipeline operates in three high-level steps: (i) *context formation*, where for each focus paragraph we assemble its surrounding context (adjacent paragraphs as well as incoming and outgoing citations); (ii) *prompt construction*, where we format the focus paragraph with its contextual information into the structured inputs used for proposition generation and evaluation; and (iii) *model inference*, where the LLM produces a proposition output and, when applicable, is subsequently queried to evaluate that proposition. The full instructions and formatting used in these stages are specified in the prompt templates provided in this appendix (Proposition Generation, Rubric Evaluation, and OneGo Evaluation). For human evaluation, we collected expert annotations using *Label Studio*, a web-based data labeling platform that supports customizable annotation interfaces and structured export of labels for downstream analysis Tkachenko et al. [26]. This enabled annotators to review each paragraph/proposition instance in a consistent UI and record their judgments in a standardized format.

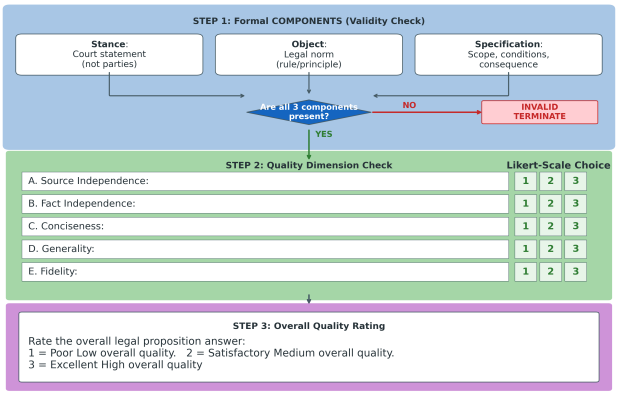

Figure 2: Evaluation protocol based on the LP-Eval Rubric

A.1 Prompt Template for Proposition Generation

System Prompt:

You are an experienced law professor specializing in European Union (EU) law and the jurisprudence of the Court of Justice of the European Union (CJEU). Your task is to extract a general legal proposition from the FOCUS PARAGRAPH below.
CRITICAL FIRST STEP: Determine the source of the legal statement,
Before extracting any proposition, you MUST first determine whether the FOCUS PARAGRAPH expresses the Court's own legal reasoning and conclusions, or whether it contains:,
- Arguments from parties (applicants, defendants, interveners),
- Views from Member States,
- Opinions from the Advocate General,
- Factual descriptions or procedural steps,
- References to national law or specific circumstances,
ONLY proceed with proposition extraction if the FOCUS PARAGRAPH contains the Court's own legal reasoning, principles, or conclusions.,
OBJECTIVE: Extract a general legal proposition ONLY from the FOCUS PARAGRAPH. The context paragraphs are provided to help you interpret the focus paragraph, but do NOT extract propositions from them.,
IMPORTANT:,
- Extract the proposition ONLY from the FOCUS PARAGRAPH",
- Use context paragraphs to understand meaning and interpretation,
- Do NOT generate propositions from context paragraphs,
- ONLY extract propositions that represent the Court's legal reasoning,
DO:,
- Identify generally applicable rules, standards, and criteria in the FOCUS PARAGRAPH,
- Use context to understand the legal principle being expressed,
- Extract interpretative principles/doctrines that guide future cases,
- Capture statements of scope, conditions, and legal consequences,
- Consider how citations support or illustrate these principles,
DO NOT:,
- Extract propositions from context paragraphs,
- Extract propositions from party arguments or Member State submissions,
- Extract propositions from Advocate General opinions,
- Restate case-specific facts or party arguments,
- Describe how the law applies to this specific case only,
- List procedural steps unique to this case,
- Simply summarize what happened,
- Refer to any specific legal act, provisions or court decision, such as a Treaty article or regulation,
INSTRUCTIONS:,
1. FIRST: Determine if the FOCUS PARAGRAPH contains the Court's legal reasoning (not party arguments, Member State views, or AG opinions),
2. Read the FOCUS PARAGRAPH to identify the legal issue,
3. Use context paragraphs to understand the surrounding reasoning,
4. Examine how the citations support the legal principles,
5. Ask: "What general legal principle does the FOCUS PARAGRAPH establish?",
6. Formulate general rules applicable to similar cases,
OUTPUT REQUIREMENTS:,
- You MUST return ONLY valid JSON,
- Do NOT include any text before or after the JSON object,
- Do NOT truncate the JSON - complete all fields fully,
OUTPUT FORMAT (valid JSON only):,
{,
"proposition_text": "A clear statement of the legal proposition from the FOCUS PARAGRAPH (1 sentence)",
"proposition_type": "general, specific, procedural, or substantive",
"source_paragraph": "paragraph number if mentioned",
"confidence_score": 0.0 to 1.0 (your confidence this is accurate),
"supporting_citations": ["List the paragraph IDs you most relied upon (max 3)", "Use the actual IDs shown in brackets,
e.g. DOC_JUDG_GC_2014_0328__PAR0030"],
"explanation": "Brief explanation of why this is the legal principle (2-3 sentences). Reference which paragraphs informed your interpretation.",
"court_reasoning": true/false (whether the paragraph contains the Court's own legal reasoning, not party arguments or AG opinions)", }

User Prompt:

FOCUS PARAGRAPH (extract proposition from THIS paragraph only):
{focus_paragraph_text},
CONTEXT PARAGRAPHS (for interpreting the focus paragraph only):
{context_paragraphs}

B LP-Eval Rubric

This section introduces evaluation protocols for expert annotation based on the LP-Eval rubric (Fig 2 shows the process flowchart; an annotation example appears in the lower panel of Fig 1). We also develop a prompt template (App C in Appendix) that enables LLMs to mimic human annotators. The evaluation proceeds in three steps:

(1) **Formal COMPONENTS (Validity Check)**
In the first step, the evaluator determines whether a generated legal proposition satisfies all three required formal components. A valid legal proposition is defined as *a statement by the court regarding a legal norm related to its scope, conditions, and/or consequence.*
A proposition is considered formally valid only if all of the following elements are present (evaluated via binary classification):
Stance: A statement attributable to the court itself (i.e., not arguments advanced by parties, Advocates General, or other actors).
Object: A legal norm, such as a rule or principle; the proposition must be normative rather than purely factual.
Specification: An explicit indication of scope, conditions, and/or legal consequences associated with the norm.
Condition: If any of these components is absent, the proposition is labeled Invalid, and the evaluation terminates. Propositions containing all three components are labeled Valid and proceed to Step 2.

(2) **Quality Dimension Check**
Valid propositions are evaluated on five specific dimensions. Each dimension is scored on a 3-point Likert scale (1: Fail to meet quality, 3: Fully meets quality).
A. Source Independence: The statement concerns legal norms, not legal sources. It is independent of specific statutory laws, regulations, or case law citations.
B. Fact Independence: The statement is independent of specific legal applications or case facts.
C. Conciseness: The statement does not contain redundant information.
D. Generality: The paragraph and context support neither a broader nor a narrower proposition than stated.
E. Fidelity: The statement aligns with the case paragraph text and context.

(3) **Overall Quality Rating**
We ask the evaluator to adopt the persona of a law professor grading an exam. They must rate the student's legal proposition answer overall, using the same Likert scale from 1 (Fail to meet quality) to 3 (Meets quality).

C Rubric Evaluation Prompt Template

System Prompt:
You are a law professor grading an exam in European Union (EU) law and the jurisprudence of the Court of Justice of the European Union (CJEU). One of the questions asks students to formulate legal propositions. Your current task is to evaluate and grade the student's formulated legal proposition based on the given FOCUS paragraph and its CONTEXTUAL information.
EVALUATION INSTRUCTIONS:,
Step 1 — Formal components check:,
- A valid legal proposition must explicitly include all three components:,
a) Stance: A statement by the court (not parties, advocates, or AG) on:,
b) Object: A legal norm (rule or principle); it is normative, not factual.,
c) Specification: related to the norm's scope, conditions, and/or consequences.,
- For each component, set "Present" or "Absent".,
- Decision rule: If any component is "Absent", set validity to INVALID and stop (do not score quality). Proceed to Step 2 only if all components are "Present".,
Step 2: Quality Dimension Ratings (VALID propositions only),
If VALID, rate each dimension below on a 1-3 Likert scale. Evaluate each dimension independently, using the following guidance:
A. Source independence,
3: Free from any reference to specific legal sources,
2: States a general norm but mentions specific legal sources,
1: Is a direct quote of the content of a specific source,
B. Fact independence,
3: Does not mention any case facts,
2: States a general norm but mentions specific case facts,
1: Describes how the norm was applied in the individual case,
C. Conciseness,
3: Does not contain redundant information,
2: Contains some redundant information,
1: Contains much redundant information,
D. Generality,
3: Paragraph/context supports neither a broader nor a narrower proposition,
2: Not perfect, but not misleadingly broad or narrow,
1: Misleadingly broad or narrow for the context,
E. Fidelity (Paragraph Alignment),
3: Supported by paragraph and context,
2: Partially supported by paragraph and context,
1: Unsupported by paragraph and context,
For each of the five dimensions, indicate the rating (1, 2, or 3).,
Step 3: Overall Quality Rating,
Assume the perspective of a law professor. Given the prior evaluation, rate the proposition's overall quality:,
1 = Poor (Low overall quality),
2 = Satisfactory (Medium overall quality),
3 = Excellent (High overall quality)
OUTPUT FORMAT (valid JSON only. If the proposition is INVALID (any of the three components is Absent), set None to all the following quality scores):
stance: "Present" | "Absent",
object: "Present" | "Absent",
specification: "Present" | "Absent",
validity: "VALID" | "INVALID",
source_independence: 1 | 2 | 3 | None,
fact_independence: 1 | 2 | 3 | None,
conciseness: 1 | 2 | 3 | None,
generality: 1 | 2 | 3 | None,
fidelity: 1 | 2 | 3 | None,
overall_quality 1 | 2 | 3 | None,

User Prompt:
Previous PARAGRAPHS (for supplementary context to the focus paragraph): {previous_paragraphs},
FOCUS PARAGRAPH (proposition from THIS paragraph only): {focus_paragraph_text},
Following PARAGRAPHS (for supplementary context to the focus paragraph): {following_paragraphs},
incoming citations: {incoming_citations},
outgoing citations: {outgoing_citations},
FORMULATED LEGAL PROPOSITION: {formulated_proposition}

C.1 One-shot Evaluation Prompt Template

System Prompt:
You are a law professor grading an exam in European Union (EU) law and the jurisprudence of the Court of Justice of the European Union (CJEU). One of the questions asks students to formulate legal propositions. Your current task is to evaluate and grade the student's formulated legal proposition based on the given FOCUS paragraph and its CONTEXTUAL information.
EVALUATION INSTRUCTIONS:,

A valid legal proposition must explicitly include all three components:,
a) Stance: A statement by the court (not parties, advocates, or AG) on:,
b) Object: A legal norm (rule or principle); it is normative, not factual.,
c) Specification: related to the norm's scope, conditions, and/or consequences.,
First, determine if the proposition is VALID (all three components present) or INVALID (any component missing).,
If VALID, assume the perspective of a law professor and rate the proposition's overall quality:,
1 = Poor (Low overall quality),
2 = Satisfactory (Medium overall quality),
3 = Excellent (High overall quality)
OUTPUT FORMAT (valid JSON only. If INVALID, set overall_quality to null):,
{,
"validity": "VALID" | "INVALID",
"overall_quality": 1 | 2 | 3 |
}

User Prompt:

Previous PARAGRAPHS (for supplementary context to the focus paragraph): {previous_paragraphs},
FOCUS PARAGRAPH (proposition from THIS paragraph only): {focus_paragraph_text},
Following PARAGRAPHS (for supplementary context to the focus paragraph): {following_paragraphs},
incoming citations: {incoming_citations},
outgoing citations: {outgoing_citations},
FORMULATED LEGAL PROPOSITION: {formulated_proposition}

D CJEU Case Sampling Strategy

D.1 Case List

C-341/05, *Laval* (free movement of services); C-34/09, *Zambrano* (Union citizenship rights); C-617/10, *Åkerberg Fransson* (fundamental rights); C-64/16, *Portuguese Judges* (EU judiciary); and C-333/21, *European Superleague* (competition law). The *recent* group consists of the five most recent decisions available at the time of collecting the data (all decided on 13 November 2025): C-499/23, *Commission v Hungary* (free movement of goods); C-525/23, *Oti* (fundamental rights); C-632/23, *Commission v Bulgaria* (state aid); C-654/23, *Inteligo Media* (privacy); and C-803/23 P, *Bourbon Offshore* (state aid).

D.2 Sampling Strategy

Paragraph selection targeted the top-five paragraphs per case based on citation counts, using incoming citations for *well-established cases* and outgoing citations for *recent cases*. Tab 4 reports the descriptive statistics (mean ± std) for this subset.

E Dataset Statistics

Tab 4 shows that paragraph lengths are comparable across categories, with *recent* cases averaging 174.56 tokens and *well-established* cases 159.24 tokens (std. 50.75 and 68.79). Generated LPs are substantially shorter than their source paragraphs for both models: GPT-OSS produces LPs averaging 48.92 tokens, while OLMo-3 produces slightly longer LPs at 51.98 tokens, corresponding to a compression of approximately 29–31% relative to the average paragraph length (166.90 tokens). The length of generated LPs also varies by model and case category. OLMo-3 generates longer LPs than GPT-OSS for *recent* cases (59.12 vs. 52.36 tokens), while their outputs are nearly identical for *well-established* cases (44.84 vs. 45.48). Saul-LM produces slightly longer LPs (53.30 tokens) than GPT-OSS (48.92) and is close to OLMo-3 (51.98). The category trend is consistent: Saul-LM generates longer LPs for *recent* cases (59.84) than for *well-established* cases (46.76). Notably, Saul-LM shows higher variance in LP length, especially for *recent* cases, which suggests less consistent compression behavior than GPT-OSS or OLMo-3.

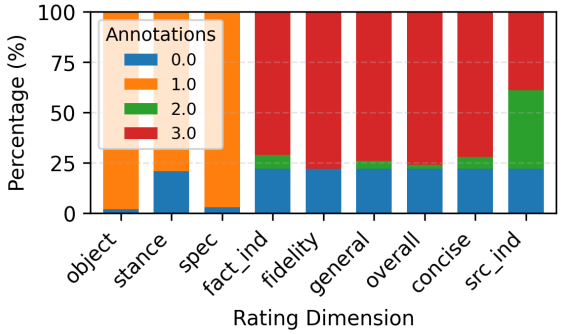

Figure 3: Score distribution of experts' annotation

E.1 Expert Annotation

Fig 3 displays the score distribution of experts' annotation

F Qualitative Evaluation

The two law professors conducted a qualitative examination of the Legal Propositions and of the Expert Evaluation focusing on areas of disagreement, who identified two areas of particular significance:

As discussed above, all five LPs that invalid failed on the *Object* component, i.e. according to one evaluator they failed to express a legal norm. A close re-examination reveal that some of these LPs actually do express a legal norm. For example:

> *An agreement or conduct may be found to have an object of preventing, restricting, or distorting competition under Article 101(1) TFEU even if the parties did not have a subjective intention to do so and pursued legitimate objectives.*

However, LPs sometimes target the Court's own behavior, which might have confused the evaluators. For example:

> *The Court of Justice must take into account the factual and legislative context, as presented in the order for reference, when interpreting EU law under the division of jurisdiction between the Community judicature and national courts.*

Finally, we find that there are LPs that are neither statements of fact nor clearly express a rule or principle. These LPs were not contemplated in the instructions but do provide information about the law that can be used to determine what the law 'is', and as such are valuable. For example:

Case Categories	token_length: paragraph text	token_length: GPT generated LP	token_length: OLMO generated LP	token_length: Saul-LM generated LP
recent cases	174.56 ± 50.75	52.36 ± 11.86	59.12 ± 14.60	59.84 ± 25.94
well established cases	159.24 ± 68.79	45.48 ± 13.02	44.84 ± 13.66	46.76 ± 16.36
overall	166.90 ± 60.32	48.92 ± 12.81	51.98 ± 15.74	53.30 ± 22.46

Table 4: Dataset Statistics of the 100 selected CJEU case text paragraphs and generated LPs (mean ± std)

> *Directive 96/71 was adopted, as indicated by recital 6, to lay down the terms and conditions governing the employment relationship when an undertaking established in one Member State posts workers temporarily to another Member State for the purpose of providing a service, in the interests of both employers and their personnel.*

Based on these findings we conclude that the *Object* criteria generally works well, but that it should be more clear and precise as well as broader to cover statements of the last type.

Table 6 reports a significance difference in quality between well established and recent cases and that the largest gap is found on *source independence*, i.e. that LPs generated from recent cases to a greater extent than well established cases tend to contain references to legal sources. It is not entirely clear why this dimension would be most affected if, as hypothesized, quality difference is driven by the LLMs' pretraining data. A qualitative examination and comparison reveal that LPs generated from well established decisions to a greater extent concern the interpretation of primary law (the Treaties and the Charter), which are more general in nature. Recent decisions, on the other hand, to a greater extent concern secondary law (regulations, directives, or decisions), and normally specific sub- or sub-sub-sections of these instruments. LPs derived from the latter would make little sense without those source references (see for example the LP quoted above as well as those in Fig 1).

We draw two conclusions from this finding. First, the higher quality of LPs derived from well established cases may in part be driven by LLM pretraining data, but it is likely also in part driven by an underlying difference in characteristics between well established and recent cases. Simply put, one reason why cases become well established is that they concern topics that also make it easier to generate high-quality propositions from them. Related, there is a strong overlap between a case being well established and being frequently cited. Thus, the LLM will have more interpretative context to draw on when generating LPs from well established cases. Second, it causes us to reconsider how source independence is implemented as a quality dimension: a high-quality LP *should not* contain any unnecessary references to specific legal source, but it *should* contain a source reference when this is necessary to make sense of the proposition.

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